算法设计 2024 年期中考试

英文原文与中文译文

考试信息

- 时间：2024 年 11 月 21 日 19:20 – 21:50，150 分钟。
- 总分 120 分；超过 100 分者按 100 分计。
- 只允许携带一张双面 A4 备忘纸。

说明：本次期中考试占最终成绩的 0%，仅用作期末考试的模拟。

对写出的每个算法都必须证明其正确性；只给正确算法而不给证明，只得部分分。

原卷来源：`github.com/ContributorSiri/lib-tests`（`Algorithm/2024Midterm.md`）

第 1 题（10 分）

> Write down the **full pseudo-code** to solve the following problem:
>
> Given a set of n intervals. Find a subset S, such that the intervals in S do not intersect and, $|S|$ is maximized.
>
> The algorithm **must** run in $O(n \log n)$ time.

写出求解下述问题的**完整伪代码**：

给定 n 个区间，求一个子集 S，使 S 中的区间两两不相交，且 $|S|$ 最大。

算法**必须**运行在 $O(n \log n)$ 时间内。

第 2 题（15 分）

> Design an algorithm to solve the following problem:
>
> Given a tree T having n nodes. Find a vertex $v \in T$ such that, if we delete v and all the edges adjacent to v from T, every connected component of the resulting forest has size $\leq n/2$.
>
> Your algorithm **must** run in $O(n)$ time. You also need to show why such v exists.

设计算法求解下述问题：

给定一棵有 n 个结点的树 T。求一个顶点 $v \in T$，使得从 T 中删去 v 以及所有与 v 相邻的边后，所得森林的每个连通块大小都不超过 $n/2$。

算法**必须**运行在 $O(n)$ 时间内；还需说明这样的 v 为何一定存在。

第 3 题（15 分）

> Design an algorithm to solve the following problem:
>
> Given a sorted array of positive integers $(a_1 \leq a_2 \leq \cdots \leq a_n)$ (they are already in the memory, so we don't need to read them), such that $a_n \leq m$. Find $b_1, b_2, \cdots, b_m$ such that, for each $1 \leq j \leq m$, j appears b_j times in array a.
>
> Your algorithm **must** run in $O(m \log n)$ time.

设计算法求解下述问题：

给定一个已排好序的正整数数组 $(a_1 \leq a_2 \leq \cdots \leq a_n)$（已在内存中，无需读入），且 $a_n \leq m$。求 $b_1, b_2, \cdots, b_m$，使得对每个 $1 \leq j \leq m$，b_j 等于 j 在数组 a 中出现的次数。

算法**必须**运行在 $O(m \log n)$ 时间内。

第 4 题（15 分）

> The description of the Max2Lin problem is as follows:
>
> n, m, k are positive integers. There are n variables $x_1, x_2, \cdots, x_n \in \mathbb{F}_2$, and m equations in $\mathbb{F}_2$. Each equation is either in form $x_i = \alpha$ or $x_{i_1} + x_{i_2} = \beta$, where $i, i_1, i_2 \in \{1, 2, \cdots, n\}$ and $\alpha, \beta \in \mathbb{F}_2$. Determine whether there is a $(x_1, x_2, \cdots, x_n) \in \mathbb{F}_2^n$ such that **at least** k equations are satisfied.
>
> Prove that Max2Lin $\in$ NP-complete.

Max2Lin 问题的描述如下：

n, m, k 为正整数。有 n 个变量 $x_1, x_2, \cdots, x_n \in \mathbb{F}_2$，以及 $\mathbb{F}_2$ 上的 m 个方程。每个方程或形如 $x_i = \alpha$，或形如 $x_{i_1} + x_{i_2} = \beta$，其中 $i, i_1, i_2 \in \{1, 2, \cdots, n\}$，$\alpha, \beta \in \mathbb{F}_2$。判定是否存在 $(x_1, x_2, \cdots, x_n) \in \mathbb{F}_2^n$，使**至少** k 个方程被满足。

证明 Max2Lin 是 NP-完全的。

第 5 题（15 分）

> Design a poly-time **constant-ratio** approximation algorithm for the following problem:
>
> Given an undirected graph G, each vertex v having degree ≤ 10 and a weight $w_v \geq 0$. Find an **independent set** $I \subseteq V(G)$ such that $\sum_{v \in I} w_v$ is maximized.
>
> In other words, your algorithm must deterministically find an independent set I', such that
>
> $$\sum_{v \in I'} w_v \geq C \cdot \sum_{v \in I^\star} w_v,$$
>
> where $I^\star$ is the optimal solution.

为下述问题设计多项式时间的**常数比**近似算法：

给定无向图 G，每个顶点 v 的度数不超过 10，并带权 $w_v \geq 0$。求一个**独立集** $I \subseteq V(G)$，使 $\sum_{v \in I} w_v$ 最大。

换言之，算法必须确定性地给出一个独立集 I'，使得

$$\sum_{v \in I'} w_v \geq C \cdot \sum_{v \in I^\star} w_v,$$

其中 $I^\star$ 为最优解。

第 6 题（15 分）

> Alice and Bob play a game on a pyramid of integers: the pyramid has n levels, level i ($1 \leq i \leq n$) having i integers. Alice and Bob both start from the only number at level 1; in each step, they need to travel to the next level by going to an adjacent number (can be the left or the right one). Finally, after $n-1$ steps, they need to reach the same number at level n. They count their score by adding every number they had visited (if both Alice and Bob visited a number, only count for once!). They want to design a strategy to maximize their score.
>
> For instance, the following graph represents a case when $n = 4$. In fact, their best strategy is: $1 \to 2 \to 5 \to 9$ for Alice, $1 \to 3 \to 6 \to 9$ for Bob. Then, their score will be $1 + 2 + 3 + 5 + 6 + 9 = 26$.
>
> Design a poly-time algorithm to solve this problem.

Alice 与 Bob 在一个整数数塔上做游戏：塔有 n 层，第 i 层（$1 \leq i \leq n$）有 i 个整数。两人都从第 1 层唯一的那个数出发；每走一步都要前往下一层中与之相邻的数（左边或右边）。经过 $n-1$ 步后，两人必须到达第 n 层的同一个数。他们的得分记为所有经过的数之和（若某个数两人都经过，只计一次！）。他们希望设计策略使得分最大。

例如，下图是 $n = 4$ 的情形。事实上他们的最优策略是：Alice 走 $1 \to 2 \to 5 \to 9$，Bob 走 $1 \to 3 \to 6 \to 9$，得分为

$$1 + 2 + 3 + 5 + 6 + 9 = 26.$$

$$\begin{array}{cccc} & & 1 & \\ & 2 & & 3 \\ 4 & & 5 & & 6 \\ 7 & 8 & 9 & 10 \end{array}$$

设计多项式时间算法求解该问题。

第 7 题（15 分）

> Suppose b is a fixed positive integer. Find a poly-time b-**approximation** algorithm for the following problem:
>
> A is a set of size n, each element $x \in A$ has a weight $w_x \geq 0$. Given m subsets $B_1, \cdots, B_m \subseteq A$, such that for each $1 \leq i \leq m$, $|B_i| \leq b$. Call $H \subseteq A$ a **hitting set**, if for each $1 \leq i \leq m$, $B_i \cap H \neq \emptyset$. Find a hitting set $H^\star$ such that $\sum_{x \in H^\star} w_x$ is minimized.

设 b 为固定的正整数。为下述问题求一个多项式时间的b-**近似**算法：

A 是大小为 n 的集合，每个元素 $x \in A$ 带权 $w_x \geq 0$。给定 m 个子集 $B_1, \cdots, B_m \subseteq A$，且对每个 $1 \leq i \leq m$ 有 $|B_i| \leq b$。称 $H \subseteq A$ 为**命中集**，若对每个 $1 \leq i \leq m$ 都有 $B_i \cap H \neq \emptyset$。求命中集 $H^\star$，使 $\sum_{x \in H^\star} w_x$ 最小。

第 8 题（20 分）

> Suppose you want to trade a kind of stock in n days. In day i ($1 \leq i \leq n$), the price of this stock is $p_i > 0$. In each day, you can either choose to buy some stocks, sell some stocks you have already bought in previous days, or do nothing. However, every stock you buy will cost you another $c_b > 0$ money; every stock you sell will cost you another $c_s > 0$ money. You need to maximize your profit after n days.
>
> **(1,** 10 **pts)** Suppose now you have **infinite** amount of money, you can only buy / sell **at most one** stock in each day, and you can hold **at most one** stock at any time. Design an algorithm that runs in $O(n)$ time.
>
> **(2,** 10 **pts)** You have M money initially, and you **cannot borrow money** (that means, you can only buy a stock if you have enough money). The amount of stocks you buy or sell in each day is **unlimited**, and the number of shares may be any **real number** (fractional shares are allowed). Design an algorithm that runs in $O(n)$ time.

设想你要在 n 天内交易某种股票。第 i 天（$1 \leq i \leq n$）该股票的价格为 $p_i > 0$。每天你可以选择买入若干股、卖出若干股（此前买入的），或者什么都不做。但你每买入一股要另付 $c_b > 0$，每卖出一股要另付 $c_s > 0$。要求 n 天之后收益最大。

(1，10 分) 现在你有**无限**资金，每天最多只能买或卖**一股**，且任意时刻最多持有**一股**。设计一个 $O(n)$ 时间的算法。

(2，10 分) 你初始有 M 元，且**不能借钱**（即只有钱够时才能买入）。每天买卖的股数**不限**，且股数可取任意**实数**（允许小数股）。设计一个 $O(n)$ 时间的算法。

校注

- 第 4 题原文 “There n variables” 缺系动词 are；另未写明 $1 \leq k \leq m$。
- 第 5 题 “each vertex having degree ≤ 10” 指整张图的最大度不超过 10，常数 C 可以依赖 10，但不能依赖 n。
- 第 6 题两人是合作关系（同一得分），故重复经过的数只计一次；“adjacent number” 指下一层中位置相邻的两个数。
- 第 2 题所求即树的重心。
- 第 8 题：(1) 已补「任意时刻最多持有一股」（原版允许持多股，堆加反悔为 $O(n \log n)$）；(2) 已补「股数取任意实数」（整数股下取整使 $O(n)$ 不成立）。两处都使全仓/空仓两状态 DP 达到 $O(n)$。